Home Assignment

Task 1 : Prepare a test strategy document based on the domain, using a chain-of-thought approach (refer to the document provided in the ZIP file)

Context : As a Senior Test Lead with sports domain you are required to generate the test strategy document.

- Ask me 5 questions one by one for creating Test Strategy
- Ask me next question after answering -
Test Strategy should follow the industry standard - Review the generated Test strategy as a Expertised test manager
Output should be a downloadable word document and review comments should be downloadable Excel format

Great—let's build this step by step like a real test strategy workshop.
✅ **Question 1 of 5**
What is the scope of the application/system under test?
(Please include details like: type of sports platform—fantasy sports, live scoring, betting, athlete management, broadcasting, etc.; key features; platforms such as web/mobile/API; and any integrations.)
👉 Once you answer, I'll move to the next question.

fantasy sports, live scoring, betting
Got it 👍
✅ **Question 2 of 5**
Who are the target users and what are the critical user journeys?
(For example: players placing bets, users creating fantasy teams, admins managing contests, live score viewers, payment flows, etc.)
👉 Please answer, and I'll move to Question 3.

players placing bets
Great, that helps narrow the focus 👍
✅ **Question 3 of 5**
What platforms and environments need to be tested?
(For example: Web, Android, iOS, APIs, third-party integrations like payment gateways, live data feeds, etc.)
👉 Share the details, and I'll move to Question 4.
Web, Android, iOS, APIs,live data feeds

Perfect 👍

✅ **Question 4 of 5**

What are the key risks and critical areas you want to prioritize in testing?
(For example: real-time scoring accuracy, betting transactions, security, performance under high traffic, regulatory compliance, etc.)

👉 Share your inputs, and I'll move to the final question.

Great—very important area 👍

✅ **Question 5 of 5**

What is your release model and testing timeline?
(For example: Agile/Scrum, sprint duration, release frequency, deadlines, regression cycles, etc.)

👉 Once you answer this final question, I'll generate:

1. 📄 A complete **Test Strategy document (Word format)**
2. 📊 A **review report (Excel format)** as an expert Test Manager

Agile/Scrum sprint duration- 2 weeks release frequency- every week

Output:

Test_Strategy.docx Test_Strategy_Review.xlsx

Task 2 : Design the automation framework structure using a chain-of-thought approach

Chat Model Used : Chat GPT

Context :

As a Senior Test Automation Engineer generate the playwright automation framework structure - Ask me 5 questions one by one for framework Creation - Ask me next question after answering - Framework should follow the industry standard - Review the generated framework as a Senior test automation lead

Great—let's design a solid, industry-standard Playwright framework step by step.

✅ **Question 1 of 5**

Which programming language do you want to use with Playwright?
(Options: TypeScript, JavaScript, Python, Java, .NET)

👉 Once you answer, I'll move to Question 2.

TypeScript,JavaScript

Good choice—both are widely used with Playwright 👍
(We'll likely standardize on **TypeScript** for better type safety, but I'll confirm based on your answers.)

✅ **Question 2 of 5**

What type of testing will the framework primarily support?
(You can choose multiple)

- UI E2E testing
- API testing
- Visual testing
- Performance testing
- Cross-browser testing

👉 Share your selection, and I'll move to Question 3.

UI E2E testing,API testing,Visual testing

Perfect—that gives us a strong, modern automation scope 👍

✅ **Question 3 of 5**
What design pattern or framework style do you prefer?
(If unsure, I can recommend one)

- Page Object Model (POM)
- Screenplay Pattern
- Hybrid (POM + utilities + services layer)
- BDD (Cucumber integration)

👉 Let me know your preference, and I'll move to Question 4.

Page Object Model (POM)

Great—POM is a solid and widely adopted choice 👍

✅ **Question 4 of 5**
What CI/CD and reporting tools do you want to integrate with the framework?
(Examples: Jenkins, GitHub Actions, Azure DevOps, Allure Reports, Playwright HTML reports, etc.)

👉 Share your preferences, and I'll move to the final question.

Azure DevOps

Nice—Azure DevOps is a solid enterprise choice 👍

✅ **Question 5 of 5**
What environments and configuration strategy do you need?
(For example: multiple environments like Dev/QA/UAT/Prod, environment-specific configs, secrets management, test data strategy, etc.)

👉 Once you answer this final question, I'll generate:

- 🧩 Complete **Playwright framework structure (industry standard)**
- 🧪 Sample code snippets (POM, API, visual testing)
- 📊 **Expert review** as a Senior Automation Lead with improvement suggestions

multiple environments like Dev/QA/UAT/Prod

Excellent—now I have everything needed to design a **production-grade Playwright framework** 👍

🧩 **Playwright Automation Framework (Industry Standard)**
🔧 **Tech Stack**

- **Language:** TypeScript (preferred over JavaScript for scalability)
- **Framework:** Playwright Test Runner
- **Design Pattern:** Page Object Model (POM)
- **CI/CD:** Azure DevOps

- **Testing Types:** UI E2E, API, Visual Testing
- **Environments:** Dev / QA / UAT / Prod

Generated Output :

week2-day1 answers.pdf